



EIMBox - DBBL Rocket Bill Payment Integration Specification

Document Version: 1.1

Target Audience: DBBL Rocket Technical Integration Team, Aggregator Developers, System Administrators

Protocol: HTTPS / REST (JSON)

Production Base URL: **<https://console.eimbox.com/api/payment/v1/>**

Standard Compliance: Dutch-Bangla Bank Limited (DBBL) Bill Payment Module Specification v1.1

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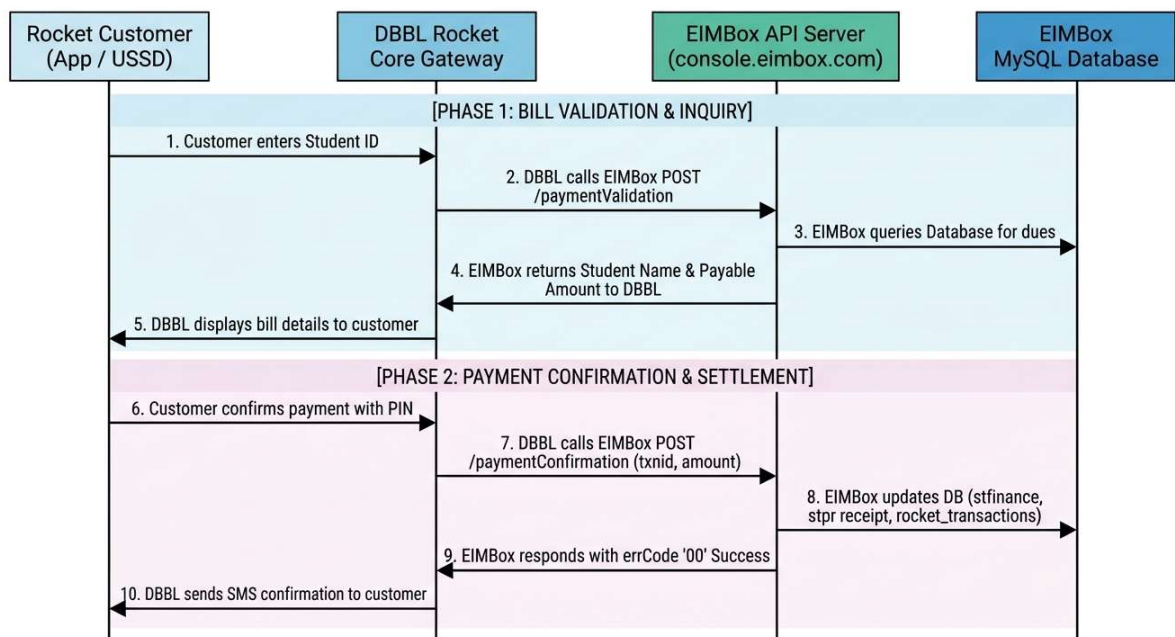
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1. Overview & Architecture

The EIMBox Educational Institution Management System provides automated student fee settlement and billing integration for Dutch-Bangla Bank Limited (DBBL) Rocket Mobile Banking.

Through this integration:

1. Guardians or students select the EIMBox Biller in their Rocket Mobile App or dial USSD (*322#).
2. The user inputs their Student ID / Reference Number.
3. Rocket queries the EIMBox Validation API, which dynamically calculates real-time unpaid dues and returns the student's name, class, section, roll, and payable amount.
4. Upon user confirmation and successful pin verification in the Rocket engine, Rocket dispatches a transaction notification to the EIMBox Confirmation API, which instantly updates the student finance ledger, issues a receipt voucher and logs transaction audit trails.
5. In case of network timeout or reconciliation, Rocket's automated batch scheduler calls the Inquiry API to verify settlement status.



2. Security & Authentication

Communication between DBBL Rocket and EIMBox must be over TLS 1.2+ (HTTPS).

2.1 HTTP Basic Authentication Header

Every HTTP request must include the Authorization header containing the Base64-encoded credentials:

```
Authorization: Basic <base64_encode(User_ID:Password)>  
Content-Type: application/json; charset=UTF-8
```

2.2 Payload Credentials

As mandated by the DBBL specification, requests also transmit user credentials directly in the JSON body:

- usrid / userid: Biller User ID
- pswrd / password: Biller Password

2.3 Authentication Models Supported:

- Global Aggregator Model (Default): All requests authenticated against central DBBL Biller partner credentials (i.e. Rocket / Rocket123).
- Multi-Tenant / Per-Institute Model: For institutions holding independent Biller IDs with DBBL, the gateway checks the payment_gateway_keys table matching biller_id, api_key, and secret_key against the specific sccode/EIIN (School Code).

3. Reference Parameters Convention

DBBL Rocket provides 3 reference fields (refNo1, refNo2, refNo3). EIMBox standard mapping:

Parameter	Data Type	Requirement	Description	Accepted Formats
refNo1	Numeric / String (10-25)	Mandatory	Student Identifier (Student's ID Number)	• 10-digit Student ID (e.g. 1031871631)
refNo2	Numeric / String (6)	Mandatory (for independent schools)	Institution EIIN / School Code (School EIIN Number)	6-digit Institution Code (e.g. 103187) fixed for each school.
refNo3	Numeric / String (4-10)	Optional	Academic Session Year	Academic Year (e.g. 2026). If empty, automatically resolves to the student's latest active academic session from record.

4. API Endpoints Specification

4.1 Validation API (`/paymentValidation`)

Validates student existence, queries real-time dues from Students Payment Ledger, and returns the student name, academic details, and exact amount payable.

- **Endpoint URL:** <https://console.eimbox.com/api/payment/v1/paymentValidation.php>
- HTTP Method: POST
- Type: Amount Provided from Partner API (DBBL Doc Section 4.2.b)

Request Fields:

Parameter	Type	Required	Description
usrid	String	Yes	Biller User ID (e.g. Rocket00)
pswrd	String	Yes	Biller Password (e.g. Rocket12321)
refNo1	String	Yes	10-digit Student ID (e.g. 1031871631)
refNo2	String	Yes	6-digit Institution Code (School Code/EIIN, e.g. 103187)
refNo3	String	No	Session Year (optional)

Sample Request:

```
{
  "usrid": "Rocket00",
  "pswrd": "Rocket12321",
  "refNo1": "1031871631",
  "refNo2": "103187",
  "refNo3": ""
}
```

Response Fields:

Parameter	Type	Description
errCode	String	00 on success; error code on failure
errMsg	String	Status description message
customerName	String	Student Full Name (e.g. Labib Shahriar)
optionalInfo1	String	Student Class and Section (e.g. Class :Ten (Science))
optionalInfo2	String	Student Roll Number and Academic Session (e.g. Roll : 2 Year: 2026)
optionalInfo3	String	Institution Code / Biller Info (e.g. Institute Sc: 102030)
amount	String	Total calculated unpaid dues (e.g. 1000)

Sample Success Response (`200 OK`):

```
{
  "errCode": "00",
  "errMsg": "Successful",
  "customerName": "MD. RAHIM AHMED",
  "optionalInfo1": "Class: Ten (A)",
  "optionalInfo2": "Roll: 01 | Year: 2026",
  "optionalInfo3": "Institute SC: 103187",
  "amount": "1200"
}
```

Sample Error Response (`200 OK`):

```
{
  "errCode": "01",
  "errMsg": "Student Not Found for given ID and Institution"
}
```

4.2 Confirmation API (`/paymentConfirmation`)

Dispatched by DBBL Rocket immediately after the user confirms payment. Settles student finance ledgers, generates receipt record with student receipt number, and logs audit entry.

- **Endpoint URL:** <https://console.eimbox.com/api/payment/v1/paymentConfirmation.php>
- HTTP Method: POST
- DBBL Doc Reference: Section 4.1.b & 4.2.c

Request Fields:

Parameter	Type	Required	Description
userid	String	Yes	Biller User ID (e.g. Rocket00)
password	String	Yes	Biller Password (e.g. Rocket12321)

Parameter	Type	Required	Description
txnid	String (15-60)	Yes	DBBL Rocket Unique Transaction ID
txndate	String (25)	No	Transaction Timestamp (e.g. 2026-09-11 13:45:00)
refno1	String	Yes	10-digit Student ID (e.g. 1031871631)
refno2	String	Yes	6-digit Institution Code (e.g. 102030)
refno3	String	No	Session Year (optional)
amount	Numeric	Yes	Paid Amount in BDT

Sample Request:

```
{
  "userid": "Rocket00",
  "password": "Rocket12321",
  "txnid": "DBBLTESTTXN012",
  "txndate": "2026-09-11 13:45:00",
  "refno1": "1031871631",
  "refno2": "103187",
  "refno3": "",
  "amount": "1200"
}
```

Idempotency & Duplicate Protection:

If Rocket sends a retry with the same txnid, EIMBox verifies existing records in rocket_transactions. If already applied, it does not deduct dues twice; instead, it immediately returns errCode: "00" with Payment Information Updated Successfully.

Sample Success Response (200 OK):

```
{
  "errCode": "00",
  "errMsg": "Payment Information Updated Successfully"
}
```


4.3 Inquiry API (`/getPaymentStatus`)

Used by the DBBL Rocket automated scheduler to verify transaction settlement status during timeouts or network drops.

- **Endpoint URL:** <https://console.eimbox.com/api/payment/v1/getPaymentStatus.php>
- HTTP Method: POST
- DBBL Doc Reference: Section 4.1.c & 4.2.d

Request Fields:

Parameter	Type	Required	Description
userid	String	Yes	Biller User ID
password	String	Yes	Biller Password
txnid	String	Yes	Transaction ID to inquire
refno1	String	No	Student Reference

Sample Request:

```
{
  "userid": "Rocket",
  "password": "Rocket123",
  "txnid": "DBBL202609110001",
  "refno1": "1001-10502"
}
```

Sample Success Response (Transaction Found):

```
{
  "errCode": "00",
  "errMsg": "Payment Information Updated Successfully"
}
```

Sample Error Response (Transaction Not Found):

```
{
  "errCode": "01",
  "errMsg": "Transaction Not Found"
}
```

5. Standard Error Codes Table (DBBL Appendix A)

Error Code	Error Message	Description / Remediation
00	Successful / Payment Information Updated Successfully	Transaction / operation completed successfully.
01	Invalid Authentication / Student Not Found	Authentication failed or student record could not be located.
02	Invalid Host Authentication	Host IP whitelisting validation failed.
05	Payment Reference Number Missing	refNo1 or School Code/EIIN parameter was not provided.
06	User ID Missing	Billor User ID header or body parameter was omitted.
07	Password Missing	Billor password was omitted.
10	Couldn't find the request body	HTTP POST body payload is empty.
11	Not a valid JSON in request body	Malformed JSON payload syntax.
12	IP Not Allowed	Origin IP is not permitted.
99	Unable to process / Invalid Parameters	Database error, invalid amount, or unhandled exception.

6. Integration Testing & Verification

6.1 Testing with cURL

Validation Test:

```
curl -i -X POST "https://console.eimbox.com/api/payment/v1/paymentValidation" \
-H "Content-Type: application/json" \
-u "Rocket00:Rocket12321" \
-d '{
  "usrid": "Rocket00",
  "pswr": "Rocket12321",
  "refNo1": "1031871631",
  "refNo2": "103187",
  "refNo3": ""
}'
```

Confirmation Test:

```
curl -i -X POST "https://console.eimbox.com/api/payment/v1/paymentConfirmation" \
-H "Content-Type: application/json" \
-u "Rocket00:Rocket12321" \
-d '{
  "userid": "Rocket00",
  "password": "Rocket12321",
  "txnid": "DBBLTESTTXN012",
  "txndate": "2026-09-11 13:45:00",
  "refno1": "1031871631",
  "refno2": "103187",
  "amount": "1200"
}'
```

Inquiry Test:

```
curl -i -X POST "https://console.eimbox.com/api/payment/v1/getPaymentStatus" \  
-H "Content-Type: application/json" \  
-u "Rocket00:Rocket12321" \  
-d '{  
  "userid": "Rocket00",  
  "password": "Rocket12321",  
  "txnid": "DBBLTESTTXN012"  
}'
```

7. Audit & Troubleshooting Logs

All transaction exchanges are recorded in a log file, rocket transaction table, payment ledger and financial audit ledger.

For technical support or sandbox whitelisting, contact the EIMBox Engineering Team at **support@eimbox.com**.